

Adaptive step size strategies for inexact first order method



What do we find?

Can Adaptive sizes remove the error floor without sacrificing efficiency?

What is this work about ?

Why does it matter?

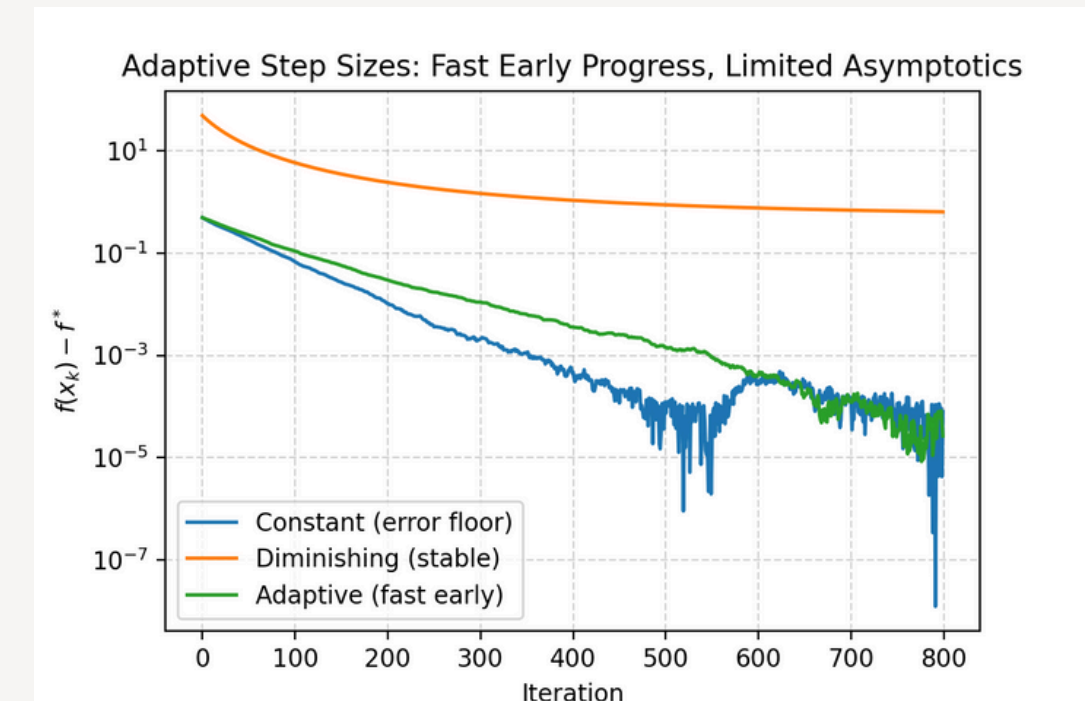
What step size do we consider?

Understanding how step size choices govern convergence behavior in first-order method with persistently inexact gradients.

Clarifies when adaptive heuristics help - and when they fundamentally cannot.

Constant, diminishing, and adaptive step size rule based on gradient variability.

(Result are supported by a convergence analysis under strong convexity)



- Constant steps are fast but stagnate.
- Diminishing steps are slow but stable.
- Adaptive steps help early, but remain noise-sensitive.